



**ENHANCING PATIENT INFORMATION MANAGEMENT WITH BIOMETRIC
REGISTRATION AND RETRIEVAL**

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Date

CERTIFICATION

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DEDICATION

I devote this thesis to my late father, Alh. Lang B. Jaiteh, may Allah SWT grant him jannatul Firdausi, and to my mother Aja Awa Ceesay for always praying for me to succeed academically.

ACKNOWLEDGEMENT

My gratitude goes to Allah S.W.T for granting me good health, protection, guidance, and helping me this far in my education. I also wish to acknowledge my family and friends for their unwavering supports, love and encouragement in helping me to complete my education. My sincerest thanks go to my hardworking supervisor Mr Kutubo Jaiteh, his constant inspiration, patience, guidance, encouragement and advice that assisted me to complete this thesis.

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ABSTRACT

Information and Communication Technology (ICT) has continued to play a crucial role in the implementation of efficient healthcare systems across the globe. Medical information is the most sensitive and important information in terms of privacy that contains all information related to the patient.

A robust and effective medical information system should be made up of storing, editable, and making available all clinical information to the doctors and staff for patient care (Kalaiselvan & Bahari, 2021). However, the privacy of patients' data has remained a major concern for most users of the Health Information System. Patient records are very vital for patient care, poorly kept or misinterpreted patient records could lead to wrong prescription or death of patient.

The integration of biometrics for identification is key in improving security and privacy within healthcare systems. The biometric system will help to eliminate the problem of inappropriate data keeping, inaccurate reports, time wastage in storing, processing and retrieving information that comes along with the traditional method of managing patient's records in order to improve the overall efficiency and performance of the hospital. The privacy and security issues in the health sector are mainly centered around user authentication, data confidentiality and patient privacy protection. Biometric technology has considerable opportunities to cope with the above security issues by providing reliable and secure user authentication. The major aim of this study is to design and implement GamCare Health Management System with fingerprint biometric for authentication. The technology allows for the accurate registration of patient fingerprints and the storage of full patient history. The doctor accesses the patients' saved information at any time (Fakhri et al., 2022).

The core functionalities of this system were designed using Laravel for the backend and Vue JavaScript (JS) for the frontend.

CHAPTER ONE

1.0 INTRODUCTION

1.1 Background of the study

The rapid growth in information and Communication Technology (ICT) has strongly influence human activities globally including the delivery of service models of many organizations with respect to healthcare, businesses, education etc. ITC has been applied in hospitals for the administration and management of patients, human resources, procurement, emergency fleet management, and much more. One of the essentials of diagnostics in modern healthcare delivery is patient medical or health records. A comprehensive healthcare system relies upon the capacity of the healthcare providers to promptly assess a patient's test outcomes, earlier treatment notes, current medicines, to mention few (Okediran et al., 2020).

The healthcare sector faces a significant number of challenges, one of which is the inefficient patient information management system.

With the prevalence of the file-folder management system in many healthcare facilities based in both rural and urban areas, the file-folder management system is easily susceptible to destruction through floods and other disasters (Parker, 2021).

The management and retrieval of patient information plays a very significant role in ensuring effective and timely medical interventions.

In numerous African nations, including the Gambia,

the traditional methods Hospitals rely on in registering patients, keeping patient data and retrieval is manual record keeping which are documented on paper forms. The files are often stored on large, disorganized shelves, retrieving them becomes challenging because it is time consuming to go through all these bulk files to retrieve a patient's medical history. This method is not only labor-intensive but can be prone to errors like incorrect data entry and lost files due to duplicate of records. Manual patient record delays in accessing a patient's medical history, incorrect or incomplete records can result in incorrect treatments being given, potentially harming patients. The amount of time medical staff spends on searching patient files can instead be spent on patient care, these delays can lead to delays in diagnosis and treatment. Additionally, these systems are vulnerable to physical damages, such as fire out break or floods

which can lead to the permanent loss of important patient data. The paper-based record systems lack confidentiality because they are shared among many users and patient privacy can be compromised. Having electronic health records has helped hospitals reduce paperwork and alleviate the shortage of healthcare workers. As more and more industries embrace the features of ICTs to enhance the way they serve their customers, the healthcare industry is largely affected because effective or ineffective customer can lead to permanent impairment or death (Otuagoma et al., 2023).

It is essential to use trusted technology for storing and retrieving records that allow users to quickly authenticate themselves.

The introduction of biometric technology in the healthcare system has the potential of transforming patient information management. Replacing the traditional manual record system with biometric registration and retrieval can optimize operations in the healthcare sector, improve data protection and improve the speed and accuracy of patient care. By associating patient data with unique biometric identifiers, healthcare providers can efficiently access and update medical records, reducing the risk of errors and improving continuity of care (Okediran et al., 2020).

The manual recording keeping, and retrieval systems are time consuming in terms of both recording information and retrieving it when needed. Thus, there need be a system that will tackle these problems.

This thesis aims to explore the potential of biometric registration and retrieval systems as a solution to the challenges above, aiming to enhance patient information management in Gambia. It will also examine the potential integration of these advanced technologies into patient biometric information systems, considering their applications and impact on healthcare delivery.

Furthermore, this research will consider the specific needs of healthcare environment including smooth integration of biometric information systems with the existing manual record keeping system. Fingerprint is thought to be the most accurate and efficient biometric identification approach is thought to be a fingerprint. Everybody has a different fingerprint, and they do not alter overtime. Users of this system will always have access to healthcare-related information because their fingerprint will be associated with their medical records (Kumar et al.,2020).

1.2 STATEMENT OF RESEARCH PROBLEMS

The inefficiency of manual patient information management systems can have serious implications for patient safety and overall quality of healthcare.

- **Inaccuracy and Errors:** The traditional manual patient information management techniques are still widely used in hospitals throughout the country. This approach is prone to errors as handwritten records may not always be readable or visible and they tend to wear out easily. Missing information in medical records can lead to misdiagnoses, wrong prescription and even death.
- **Inefficiency and Time Consuming:** The process of manually recording, keeping, and retrieving patient information is very tedious as it is time consuming in locating, accessing and verifying patient files as they are piled up in bulks. and. The amount of time medical staff spends on searching patient files can be spent on patient care. Most patients complain about the time spent between walking into the hospital and being attended by hospital staff, especially doctors (Kyambille & Kalegele, 2015). This can result in long waiting times for patients, delay or postponement of clinical decision-making and increase administrative workload for healthcare workers.
- **Data Privacy and Protection:** The physical documents are at a risk of being lost, damaged, and unauthorized access to storage facilities. Healthcare records are essential to every patient and medical experts as well. There should be a system that only authorized person gains access to the patient record, it can lead to the wrong medication or even the death of the patient (Fatima et al., 2019). Natural disasters such as floods or fire and mishandling of records can lead to permanent loss of critical patient information.
- **limited Access to patient data:** The traditional systems limit the ability of healthcare providers to promptly retrieve patient data during emergency situations. This makes it difficult for healthcare providers to access patient information quickly, especially in emergency situations.

This may disturb the flow of healthcare and restrict the availability of important patient information for making well-informed decisions on the treatment-on-treatment plan (Thrasher & Revels, 2012).

- **Costs and Operations:** Maintaining manual patient information systems involves significant financial costs pertaining to paper supplies, storage infrastructure and administrative tasks. The inefficient data retrieval and management further increased operational cost. These financial burdens do not lead to improved service delivery in the healthcare sector.
- **Insufficient Technological Infrastructures:** Frequent power interruptions are a significant obstacle to the advancement of digitalization in the country. This can lead to damage to digital/ICT equipment, and the expense of operating on standby generators is often expensive to manage.

1.3 OBJECTIVE OF STUDY

- The main objective of this research is to examine the current practices of managing patient information in hospitals. This involves monitoring the methods used for recording, storing, and retrieving patients and coming up with solutions to mitigate these challenges to optimize healthcare operations.
- To Analyze the impact of biometric systems on overall Healthcare delivery, including patient satisfaction, reduction in medical errors, and increase efficiency in healthcare delivery.
- Eliminate redundancy in terms of data storage. The biometric-based system facilitates the maintenance of health records. Patients are not required to bring any documents, prescriptions, or other items. The management of patient records is also made simpler for the practitioner (Nair, 2022).
- Increase Efficiency and Interactivity in any area of specialization in the hospital. By associating patient data with unique biometric identifiers, healthcare providers

can efficiently access and update medical records, reducing the risk of errors and improving continuity of care (Okediran et al., 2020).

- Investigate the challenges faced in handling health records in the health sector across the country.

CHAPTER TWO

2.0 LITERATURE REVIEW

The healthcare sector faces many challenges including inadequate infrastructure and insufficient funding. The administration of patient data has a big influence on the standard of care and is a crucial part of providing healthcare. Manual record-keeping methods in numerous hospitals in the country are filled with many inconsistencies with inefficiencies and inaccuracies. Biometric fingerprint registration and retrieval systems offer a hopeful remedy to these issues by providing precise and secure patient identification as well as effective data management (Guo et al., 2007).

This chapter presents the review of literature in this order: Conceptual review, empirical review, Theoretical review, and existing gap in literature.

2.1 Conceptual Review

Biometrics in simple terms is the measurement of a human being using physical and behavioral characteristics. It is a pattern of ridges and valleys on the surface of a fingertip, the formation of which is determined during the first seventh months of fetal development. The field of biometric deals with finding ways to evaluate and use a person's physical and behavioral characteristics for identification and verification (Nair, 2022).

Biometric as the identifier can play a significant role in maintaining security and privacy of healthcare system. Biometric identification is the measurable unique traits of the individual (Fatima et al., 2019).

Biometric identification systems make use of biological data, in the form, of voice, face and signature scan. The most common physical biometrics include finger-scan, retina scan, hand-scan and iris-scan.

Fingerprint recognition is one of the first and original biometric technologies that used for person identification (Jain et al., 2004). Fingerprint is a pattern on the pulp of each finger, this pattern is considered to be unique for every person. The fingerprint scanner captures the image of fingerprint, a unique pattern that is extracted from the image of the fingerprint and it matches it against the images stored in the database to establish the identity.

Fingerprint scanners have two jobs: get the image of a fingerprint and to also determine if the pattern of ridges and valleys of the image matches the pattern of ridges and valleys in pre-

scanned fingerprints. There are two main methods of getting the image of fingerprints, and they are optical scanning and capacitance scanning (Otuagoma et al., 2023).

Biometric identification cannot be forgotten or stolen and involves the use of IT to acquire, store, manage and transmit genetic data. It enables a human to be identified and authenticated through a set of recognizable and verifiable biometric data such as face, fingerprint, iris and voice data.

Biometrics make use of traits at a genetic and molecular level which may include features like DNA or blood, whilst physiological biometrics involves the individual physical traits like a fingerprint, iris or the shape of the face.

Fingerprint recognition, which measures a finger's unique pattern, is one of the oldest forms of biometric identification. Biometric fingerprint technology is a method that relies on unique physical identification of individuals based on their fingerprint.

The fingerprint based medical system is a more efficient means of storing clinical data for patients. The fingerprint scanner tool is a recognition technology, that can quickly and easily determine the patient's past medical history with a single scan. The system consists of an Arduino UNO board, a fingerprint sensor, a secure digital (SD) card module, and a micro-SD card (Fakhri et al., 2022).

The Gamcare Hospital Management System allows the use of patient's fingerprint to store and retrieve patient information. This system ensures that the person accessing or being provided with medical services is indeed the correct individual, thereby increasing the security and accuracy of patient data management.

Biometric fingerprint technology provides a highly reliable and secure method of identification and authentication as they are unique and are highly accurate. The patient Biometric information systems are gaining significance in the healthcare sector as they provide a secure and more efficient method of storing and retrieving patient data. The fingerprint is thought to be the most accurate and efficient biometric identification approach is thought to be a fingerprint. Everybody has a different fingerprint, and they do not alter overtime. Users of this system will always have access to healthcare-related information (Nair, 2022).

Biometric security is based on a unique feature of an individual's body which makes it very difficult to copy, steal or replicate a fingerprint (Thrasher & Revels, 2012).

The strength of fingerprint is that it is already a proven technology that provides high levels of accuracy. The most common types of biometric identification are face and fingerprint scans, like the technology used to unlock smartphones and mobile devices. Other types of biometric might use palm vein patterns, retina scans or voice recognition. The two main functions of biometric devices are:

- I. Verifying your identity: Checking if the person is who he says he is
- II. Recognition: To establish that you are not who you claim to be.

2.2 Patient Identification

Patient identification is the matching of a patient with their own samples, specimens, records and treatment (Lichner et al., 2008). Patient identification assures the right patient gets the right treatment. This is very important when thinking about patient safety. Patient safety and in general patient identification have received less attention in less development countries (Latham et al., 2012).

During the identification process, the biometric technology compares newly scanned data with the stored templates. Matching data points can then be used to confirm or deny the individual's identity. As the pattern of fingerprint is unique even twins who share the same DNA still have got different fingerprints and it is not changed over time of human being's life (Yoon, 2014). Biometric identification will eliminate the password abuse, removes the need to remember multiple passwords, and will maintain security standards in the healthcare environment. When applied to patients, it can dramatically reduce data-entry errors and streamline patient workflow.

The three basic patterns of fingerprint that are used to recognize a person are the arch, the loop and whorl.

Biometric authentication technology, which utilizes unique biological characteristics for identification, offers a compelling alternative to manual patient data management systems. Among the various biometric technologies available, fingerprint recognition stands out as one of the most established and widely utilized due to its accuracy, user-friendly interface, and cost-effectiveness. Integrating biometric fingerprint systems into healthcare settings has the potential to address numerous challenges associated with traditional record-keeping methods. Fingerprint-based identification systems provide several benefits that make them well-suited

for healthcare use. These include the uniqueness of fingerprints, their durability over an individual's lifetime, and their ability to quickly and accurately capture and match fingerprint data.

Biometric systems can provide a higher level of security than traditional authentication methods as they are based on unique physical or behavioral characteristics of an individual (Blanco-Gonzalo et al., 2018). Fingerprints are extremely difficult to forge and replicate, which makes them an ideal biometric modality for sensitive healthcare environments where data integrity and patient privacy are paramount (Niazy et al., 2007).

2.3 Empirical Review

The use of manual methods to record patient information is a widespread problem in numerous healthcare systems across Africa, including The Gambia. This manual approach frequently results in inefficiencies, inaccuracies, and difficulties with managing patient information. This empirical review investigates the possibility of improving patient information management in The Gambia through the implementation of biometric fingerprint registration and retrieval systems. It assesses the current manual procedures, their associated challenges, as well as the advantages and potential impact of incorporating biometric systems.

The implementation of biometric fingerprint technology for management of patient information in The Gambia signifies a notable transition aimed at improving the effectiveness and protection of healthcare data. By transitioning to biometric fingerprint-based systems for patient information management, hospitals in The Gambia can overcome the limitations of the traditional manual record-keeping, paving the way for a more streamline and effective healthcare delivery system.

Biometric systems offer several advantages over traditional manual record keeping and authentication. through biometric, all these problems can solve, staff authentication and the patient record privacy can safe from unauthorized access (Fatima et al., 2019).

They are inherently more reliable than the traditional manual record keeping as biometric traits cannot be loss or forgotten, biometric traits are difficult to copy, share, or distributed among patients and they require the person being authenticated to be present at the time and point of authentication. It is difficult to forge biometrics as every human being has a unique biometric trait, biometrics-based authentication is a powerful alternative to traditional record keeping in some instances,

biometric can be used along with passwords (or tokens) for alternatives for people with genetic factors, aging, environment or with manual workers with may have large number of cuts or bruises on their fingerprints that keep changing to enhance security offered by the authentication system.

2.4 Implementation challenges:

1. Technical challenges: For successful implementation biometric fingerprint systems, it requires robust technology infrastructure. The existing IT infrastructure may be insufficient to support such systems. Hospitals and clinics may lack the necessary hardware, such as computers, biometric scanners and servers as well as good and reliable internet connectivity. The integration of biometric systems with the existing health information system can also pose technical challenges. To address these challenges requires huge investment in technology and providing continuous technical assistance needed.

2. Logistical challenges: The logistical considerations are key in the successful implementation within healthcare systems. Ensuring all health facilities, even those situated in remote or rural areas in the country, have necessary equipment and assistance. This entails the distribution of biometric scanners, setting up a central data center and establishing reliable communication networks.

3. Financial constrain: The financial cost involve in the implementation of the biometric systems can delay the pace and scope. The purchase of computers, biometric scanners, training medical personnel on how to use the system, and modernizing IT infrastructure. Additionally, the costs for system maintenance, technical support, and data storage. Securing funding for these expenses can be challenging, especially in a country with limited resources like The Gambia. One of the many problems in Africa is that the government do not support innovative technology initiatives because with such systems in place, corruption will be minimal. Alot of countries in Africa, the government is corrupt and wastes its resources (Wall et al., 2015).

4. Limited Infrastructure: The capacity of the healthcare infrastructure to accommodate biometric systems plays a very important role in ensuring successful implementation of the biometric system. It requires a substantial upgrade to both the physical and IT infrastructure. This includes ensuring uninterrupted electricity power supply, safe data storage and the ability to manage large amounts of patient data. Poor infrastructure such as unstable power supply and poor internet connectivity is given as one of the biggest challenges in E-Health in Africa (Olu

et., 2019). A convenient working environment, internet facilities, and communication systems can help to improve the performance and productivity of health care workers using electronic facilities (Omotosho et al., 2019).

2.5 Theoretical Review

In numerous African countries, including the Gambia, the predominant method of patient information management has been manual record-keeping. This medical record-keeping involves the use of paper-based systems where patient data is written, stored, and retrieved manually.

2.6 Challenges of Manual patient information Management

Manual patient information record keeping faces several challenges, which includes:

- 1. Inefficiency and Error-prone:** manual record methods are time-consuming; it requires a significant effort from healthcare staff to record and retrieve patient information. Patients spend much time in hospitals for non-functional activities such as registration, depositing money and collecting patient information. Manual recording such as taking the readings and recording it again take time (Shaw & Govinda, 2017). This method is prone to errors such as illegible handwriting, misspellings which can lead to misdiagnoses and incorrect treatment. Documents which have been written also tend to experience wear and tear because of heavy use throughout the years or poor maintenance (Parker, 2021).
- 2. Storage Issues:** physical storage of paper records requires substantial space as the files are stored in heaps. These records are vulnerable to physical harm such as fire or floods and even termite attacks. Patient records are vital for patient care, but incomplete health records or misplaced information or mix ups with another patient's record can result in wrong medication (Mogli, 2012).
- 3. Inefficient Data Retrieval:** locating specific patient records within bulky fields can be time consuming and delay patient care and treatment decisions. This causes a problem in an emergency situation where quick access to patient records is crucial in making an informed decision. An individual in need of help but unable to move or act is stranded and helpless until the ambulance arrives, however due to lack of information pertaining to the individual, the choice of first aid could either save his/her life or kill them (Parker, 2021).

4. Security and Privacy Concern: The preservation of the security and privacy of patient data poses a challenge when dealing with physical records. The unauthorized retrieval of confidential patient information may result in privacy violations, which have legal and ethical implications. Having access to personal health records raises privacy, confidentiality and security concerns. This emphasizes the need for reliable and powerful identification. Using biometrics to identify patients can prevent duplication of medical records and detect fraud while increasing patient safety (Nair, 2022).

5. Environmental Implications: The manufacturing and discarding of paper are linked to deforestation and generation of waste. The continuity of using paper-based records not only drawbacks for healthcare facilities but also plays a role in deterioration of the environment.

2.7 Advantages of Biometric Fingerprint Registration and Retrieval

The implementation of Biometric fingerprint and retrieval in patient information management offers several advantages which are:

- **Enhanced Accuracy:** Biometric fingerprints provide a high level of accuracy in patient identification. In contrast to the traditional manual method which is prone to human errors like misspellings, illegible handwritten which can lead to misdiagnoses and incorrect medical prescriptions which can even cause death to patients. This enhanced accuracy ensures that a patient's health records are liable linked to their identity, improving the overall reliability of the patient information management process.
- **Efficient Data Retrieval:** Efficiency is important in terms of emergency situations because it helps medical staff to make informed decisions. With a single fingerprint scan, medical staff can access a complete patient medical history withing a short period of time and administer the necessary medications.
- **Improved Security and Privacy:** It reduces the risk of authorized access to sensitive patient data, thereby protecting patient privacy and complying with data protection policy. Biometric technology has considerable opportunities to cope with the above security problems by providing reliable and secure user authentication. However, due to the sensitive nature of health data, the most important challenge list toward the development of an efficient security model that can guaranty data privacy and reliability, verifying that only authorized personnel can access their corresponding health data (Jahan et al., 2028).

- **Reduction in Fraud:** Biometric can also prevent fraud, because individuals who know that such a system is in place, it is less likely for anyone to try and register more than once (Mogli, 2012). Today more hospitals and healthcare systems have migrated to electronic health records, and more health information exchanges are built to coordinate care across various hospitals' networks. Because of this rapid migration, concerns have been raised about how to effectively manage data integrity to ensure it is kept free from corruption, modification, or unauthorized access (Otuagoma et al., 2023).
- **Streamline workflow:** It reduces the administrative burden on the medical staff unlike manual record keeping whereas medical staff will go through a bunch of files just to retrieve patient pass medical history and if a file is not found in time, a new file is created for the patient. The biometric system will allow them to focus more on patient care and it will also eliminate the mistakes that come up with manual record keeping and data retrieval which poses significant drawbacks. This improved efficiency can lead to patient experience and increased staff productivity.
- **Environment friendly:** The widespread use of paper results in a decrease in productivity and efficiency. Secondly, the paper manufacturing industry is a major contributor to environment degradation, greenhouse gas emission and pollutions (Koren & Prasad, 2020). The decrease in paper utilization leads to a reduction in activities such as deforestation and waste generation that are linked to manufacturing and disposal of paper. Integrating into biometric system diminishes the dependence on manual-based patient record system.

In summary, the utilization of biometric fingerprint registration and retrieval has more advantages compared to the conventional manual record systems in terms of patient information management. The implementation of biometric will go a long way in combating fraudulent activities by enhancing security and efficiency of handling patient data, thereby minimizing errors and unauthorized access while streamlining the overall process. The biometric fingerprint system will save a significant amount of time than the manual system, patient's information which is managed by the biometric system will enhance dependability, durability, and efficiency. It provides improved speed and performance, as well as better data security.

2.8 Existing Gap in Literature

Despite the advancements in healthcare technology globally, many hospitals in Africa still depend on manual recording of patient information management. However, this manual system poses significant drawbacks. Paper records are prone to damage, loss and difficulty in accessing information in a timely manner (Koren & Prasad, 2020).

Retrieving patient information using the traditional method requires a significant amount of time and labor. Healthcare practitioners spend most of their time to navigation through physical files, a practice that can cause delays in diagnosis, treatment and the overall medical care given to patients. The biometric patient information registration and retrieval system can provide quick medical staffs access to patient information and thereby also improve patient care and optimize revenue by saving labor costs associated record retrieval, photocopying of records and other processes.

Moreover, the utilization of biometric fingerprint technology has been effectively integrated into different sectors such as national identification systems and for tracking Employees's time and attendance in both private and public sectors. With the biometric clocking system, which is the most accurate way to collect employee's time and attendance information, it verifies an employee's identity based on the characteristic of their fingerprint (Muhtahiro. Et al. 2013).

CHAPTER THREE

3.0 METHODOLOGY

3.1 Research Methodology

This research adopts a qualitative approach to investigate the current state of patient information management practice in Bundung Maternal and Child Health Hospital (BMCH) formerly Jammeh Foundation Hospital. The research was focused on the obstacles and constraints linked to manual recording and retrieval systems. The qualitative method uses a variety of methods, such as intensive interviews or in-depth analysis of historical materials, and it is concerned with a comprehensive account of some event or unit (Aspers & Corte, 2019). Healthcare personnels and the administrative staff specialists were interviewed to acquire firsthand information about some of the challenges and difficulties with the existing system. In addition to interviews, direct observation within the hospital settings to document the manual processes of patient information management. This observation method also provides an insight into how information is recorded, stored, and retrieved, highlighting the possible challenges the staff faced and the risk of patient and data security.

During both the interview and the observation approach, the existing method of patient information management are in the following ways:

First, when a patient comes into the hospital, the first meeting point is the reception where the patient buys a ticket. Patients must pay 75 dalasi for a ticket and are recorded in the day record; records are kept of every shift. The shifts are divided into three shifts: the morning shift from 8:00 to 14:00, the afternoon shift from 14:00 tot 20:00 and the night shift from 20:00 to 8:00. During the ticket purchase, you will be ask if this is your first time coming to the hospital if yes, a four-digit ID number will be generated for you, and it will be indicated in your ticket if not they will use the ID that was generated for you the last time you visited. The four-digit ID number is a unique number that is given to every patient that comes to the hospital. If a return comes to the hospital, the receptionist will ask for you national identity or your phone number to retrieve you ID number that was generated the last time you visited. They have a system that is installed in a tablet where they retrieve patient ID by either using national ID or phone number.

The ID number serves as a reference in helping to track patients, then you will be sent to the consultation room where a nurse will interrogate you and check your vitals which include the temperature, blood pressure, weight, height, etc.

From there, you will be sent to the Doctor. The Doctor will determine if there needs to be a lab test depending on test results from the nurse. If there is, you will be sent to the cashier to pay for the laboratory test and then proceed to the lab to do the necessary test. After the test results are done, you will be sent back to the doctor to analyze the test results. Then you have the record keeper with a large logbook which has the record of every patient that visited the hospital, the record keeper will ask for your ticket which bears your name, diagnosis and records them. The Doctor will prescribe the necessary medications and then send you to the pharmacy to collect your medicine.



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Serial (Registration No.):

Client SPT ID: 986-2069
2969

SP No 0003486

UNIT: DATE: 20/8/20 SEX: M ☐ F ☒ CASE: New ☐ Follow-up ☐

CLIENT NAME: Awsy Celesay AGE/DOB: 57

BP: PR: RR: TEMP: SpO2: WEIGHT:

SIGNS & SYMPTOMS:

.....

.....

.....

DURATION:

.....

.....

.....

ASSESSMENT (IMNCI):

CLASSIFICATION:

DIAGNOSIS/ IMPRESSION:

.....

FOLLOW UP IN:

TREATMENT (Rx):

☐

☐

☐

☐

☐

☐

☐

Prescriber Name/ Code: **Designation:** **Signature:**

Figure 1: The current ticket and patient information

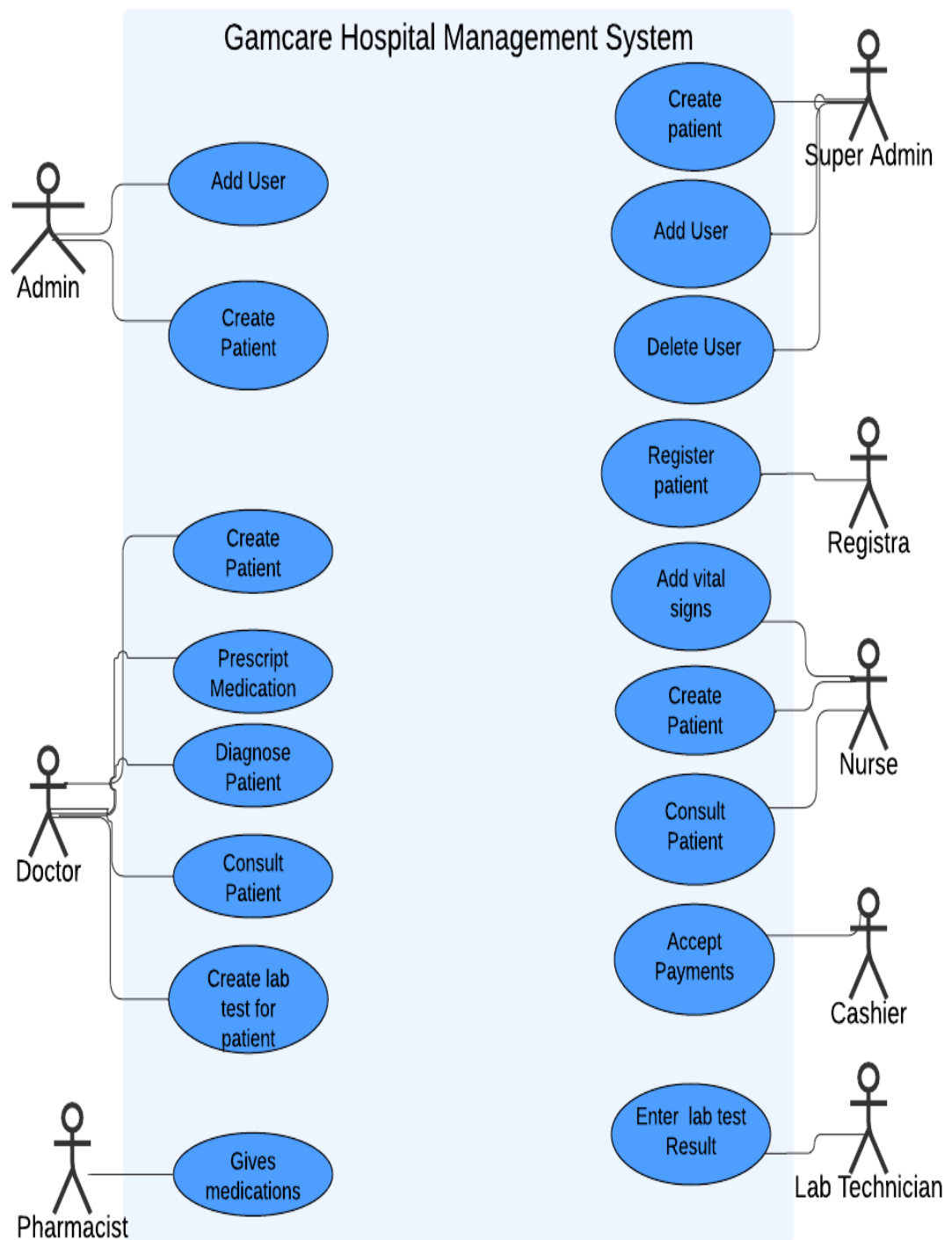


Figure2: Use case diagram of the GamCare Hospital Management System

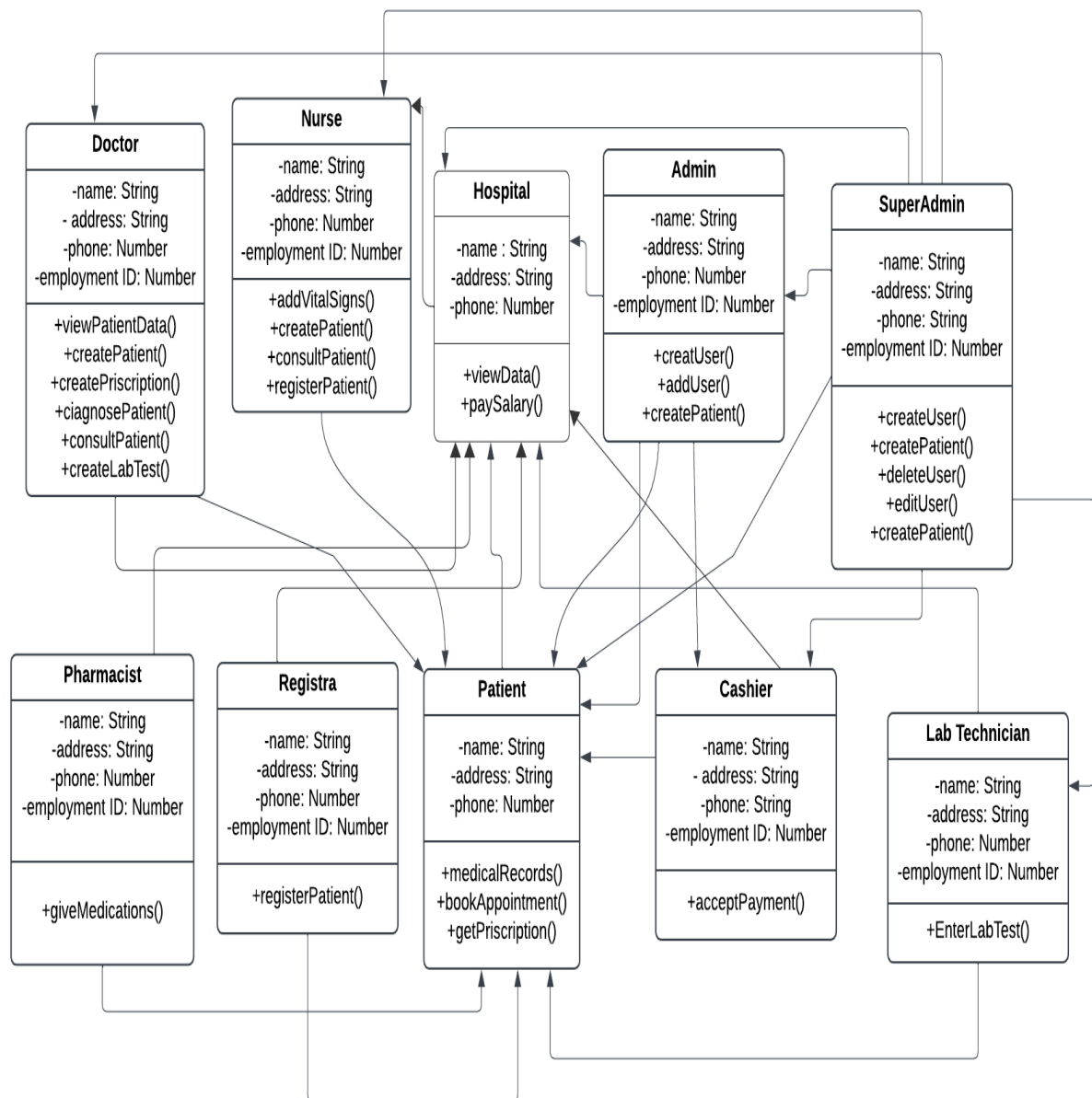


Figure 3: UML Diagram of the GamCare Hospital Management System

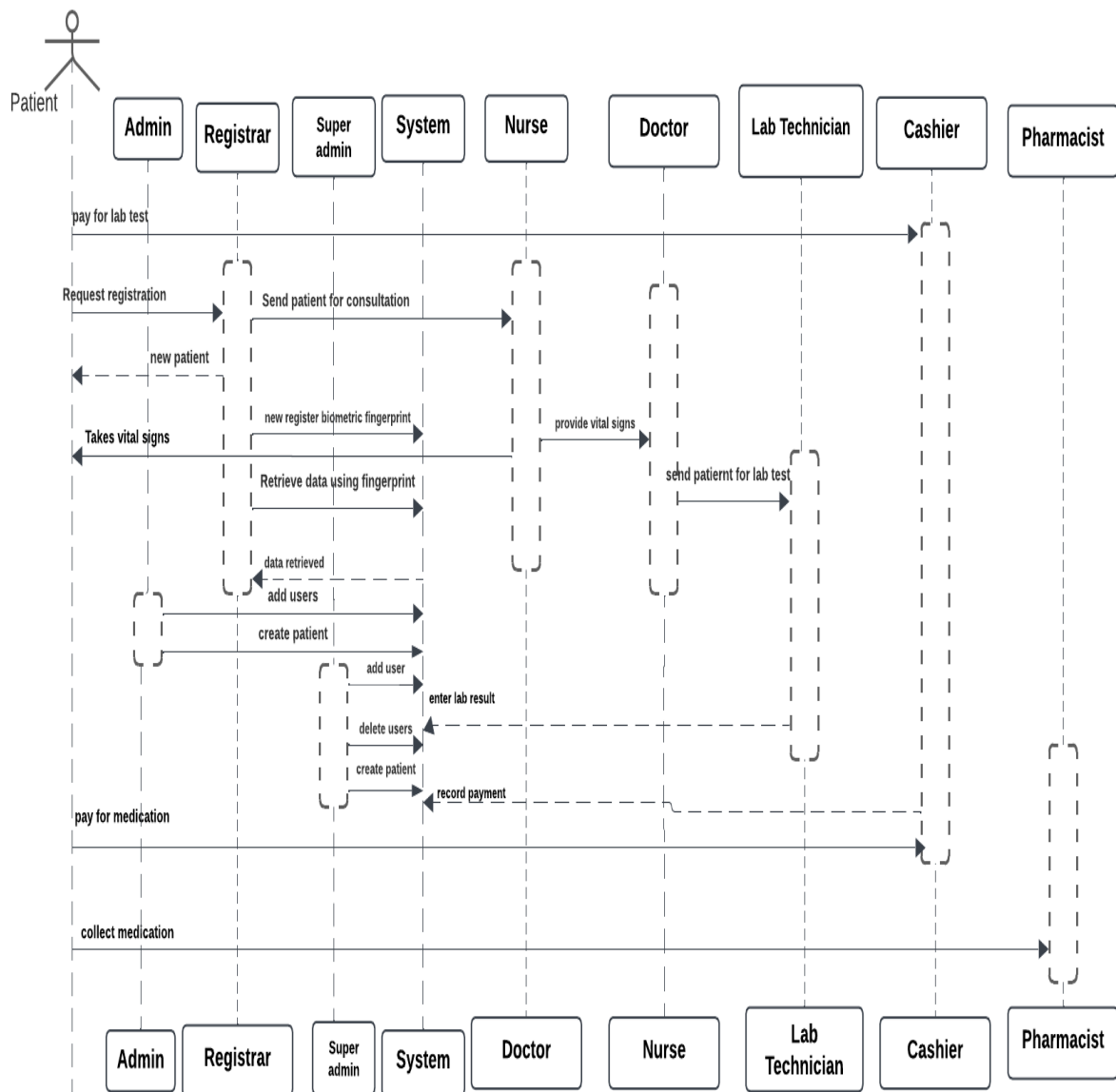


Figure 4: Sequence Diagram of the GamCare Hospital Management System

3.2 Research Design

The GamCare Hospital Management System is a software that will enable authorized hospital staff to store and / or access patients' information using their biometric fingerprint. The hospital staff is basically made up of the doctor, the nurses, the cashier, record keeper and pharmacist. Below is a detailed explanation of the steps involved:

3.3 The system consists of four modules which are:

- **The Administrator / Admin:** This module handles all the changes that are made in the database and the enrollment of users. Each user enrolled in this module is given privileges based on their status, this role is managed by IT personnel or hospital management. The admin manages users accounts, permissions, create, modify, and delete a user accounts or staff in case of firing a staff member or a staff been deceased or retired. The system has two admins, which is the super admin and admin. The super admin is given the highest authority than the admin. When you logged in the system as the admin, you can only control the users in that particular Health center which you are working for but for super admin, you can have access to all the features of the system. The super admin has the authority to audit the system, manage, monitor usage patterns and address any system failures. If a person is verified to be admin, the software opens a page different from the one opened for a regular staff because the admin does not have the privilege to add or view patient's data but can only add or delete staff. The database must contain a staff's ID, name, password, position and other data.

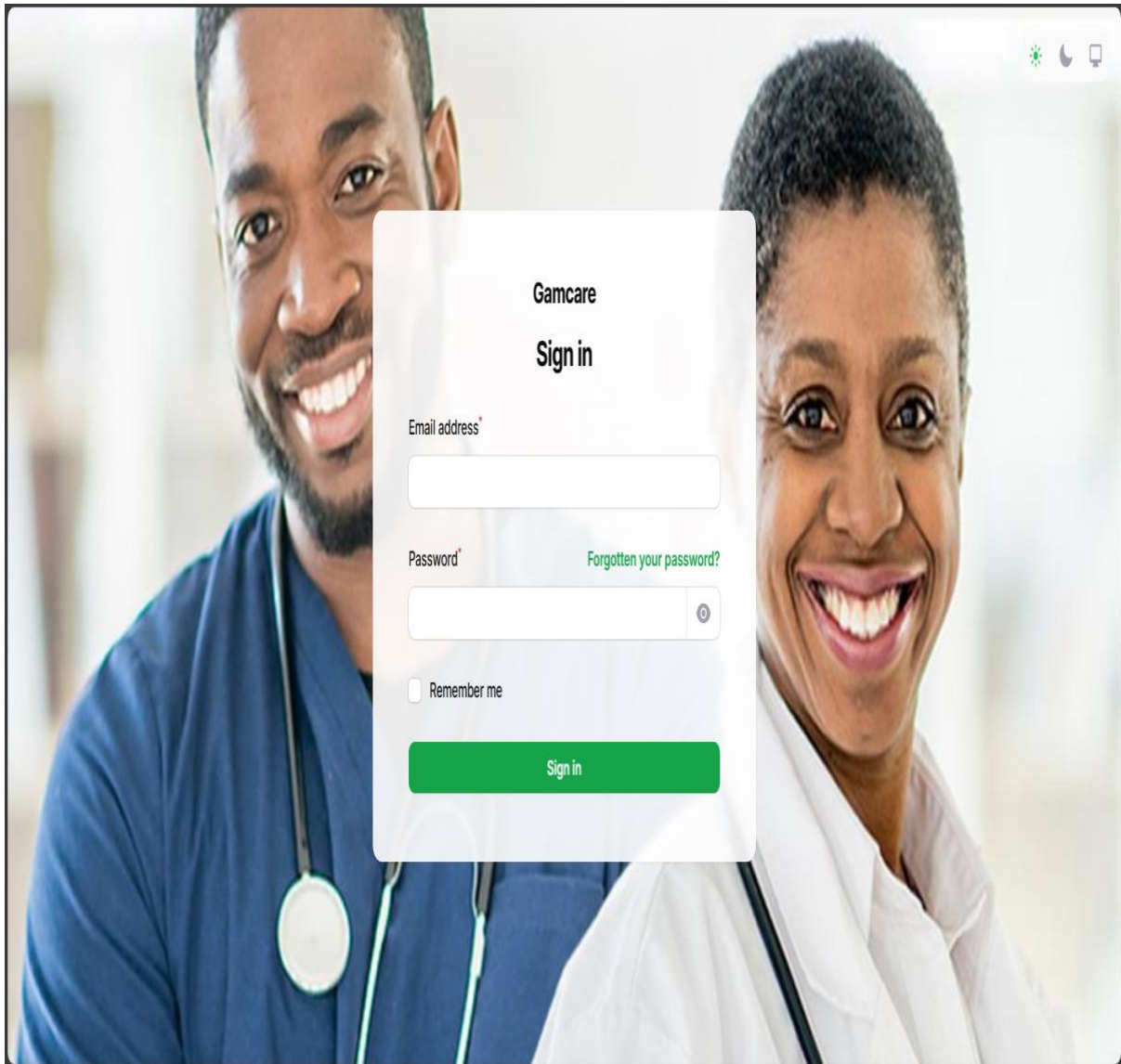


Figure 5: Login Page for the GamCare Hospital Management System.

The login page requires the Email address and Password for each staff using the system.

- **Registration:** This module handles the registration of patients in the reception and assignment of patients to a nurse. This is the first entry point in the hospital, the registration of patient is done in this module. First, the entire hospital staff will be registered into the system and will be provided with a unique ID. These will be stored in the database and will be used for authentication purposes for anyone trying to gain access to the system. During the registration of staff, it is important to state

the position of each member of staff as this will inform the system whether that particular member of staff can edit, delete, store and access it. The system takes the patients fingerprint compares the pre-scanned fingerprints in the database. If the fingerprint matches the one in the database, the system will retrieve patient data.

- **First-time visitors:** When a patient visits for the first time, the registration begins, which includes capturing the patient's fingerprint using the biometric scanner. Biometric fingerprint is scanned and all their information such as first name, surname, sex, age, address, phone number etc. Will be stored in the system; this fingerprint is then stored in the hospital's database associating it with the patient's personal and medical information.
- **Returning Patients:** For returning patients, the process is simple: the patient places their finger on the biometric scanner, allowing the system to quickly retrieve their previously stored health data. The registrar will send you to the nurse for consultation.
- **Nurse:** During the consultation, the nurse records the patient's vitals such as temperature, blood pressure (BP), weight, etc. Into the system and forward it to the doctor.
- **The Doctor:** This module allows the doctor to prescribe medications to patients. The Doctor will review the patient's vitals from the nurse and determine, if need be, for a lap test you will be sent to the cashier to pay for lap test.
- **Cashier:** All payments are made in this module. Patients must make a payment to the cashier before receiving medications or doing a lap test. After the doctor determines the need for lab test, the patient is directed to the cashier to make the necessary payments. The role of the cashier is to process payments, ensuring that the payments are accurately linked to patients record in minimizing errors and fraud. The cashier will generate a receipt to provide patients with proof of payment before proceeding to the lap.

The screenshot displays the 'Create Patient' page in the Gamcare application. The browser's address bar shows the URL '127.0.0.1:8000/patients/create'. The sidebar on the left contains the following menu items: Dashboard, Patients (highlighted), User Management (with sub-items Users, Roles, and Permissions), Settings, and Wards. The main content area is titled 'Create Patient' and contains a form with the following fields: Name (text input), Gender (dropdown menu), Address (text input), Dob (date input with format 'dd/mm/yyyy'), Age (text input), Blood group (dropdown menu), Phone No1 (text input), Phone No2 (text input), ID Number (text input with a dropdown arrow), ID Number Type (dropdown menu), Email (text input), and Fee (text input with value 'D 150'). There is also a checkbox for 'Has insurance'. To the right of the form is a 'Biometric Scan' section with a placeholder image and the text 'Please place enrolled finger on fingerprint scanner'. At the bottom of the form are three buttons: 'Create' (green), 'Create & create another' (light green), and 'Cancel' (light gray).

Figure 6: Patient registration page

The registering page is the first point of interaction in the patient management process. When patients arrive at the hospital, they are registered in the system as new patient or by retrieving their existing record using their fingerprint. The page captures patient's essential information such as name, address, age, phone number, email address and fingerprint data for identification purposes.

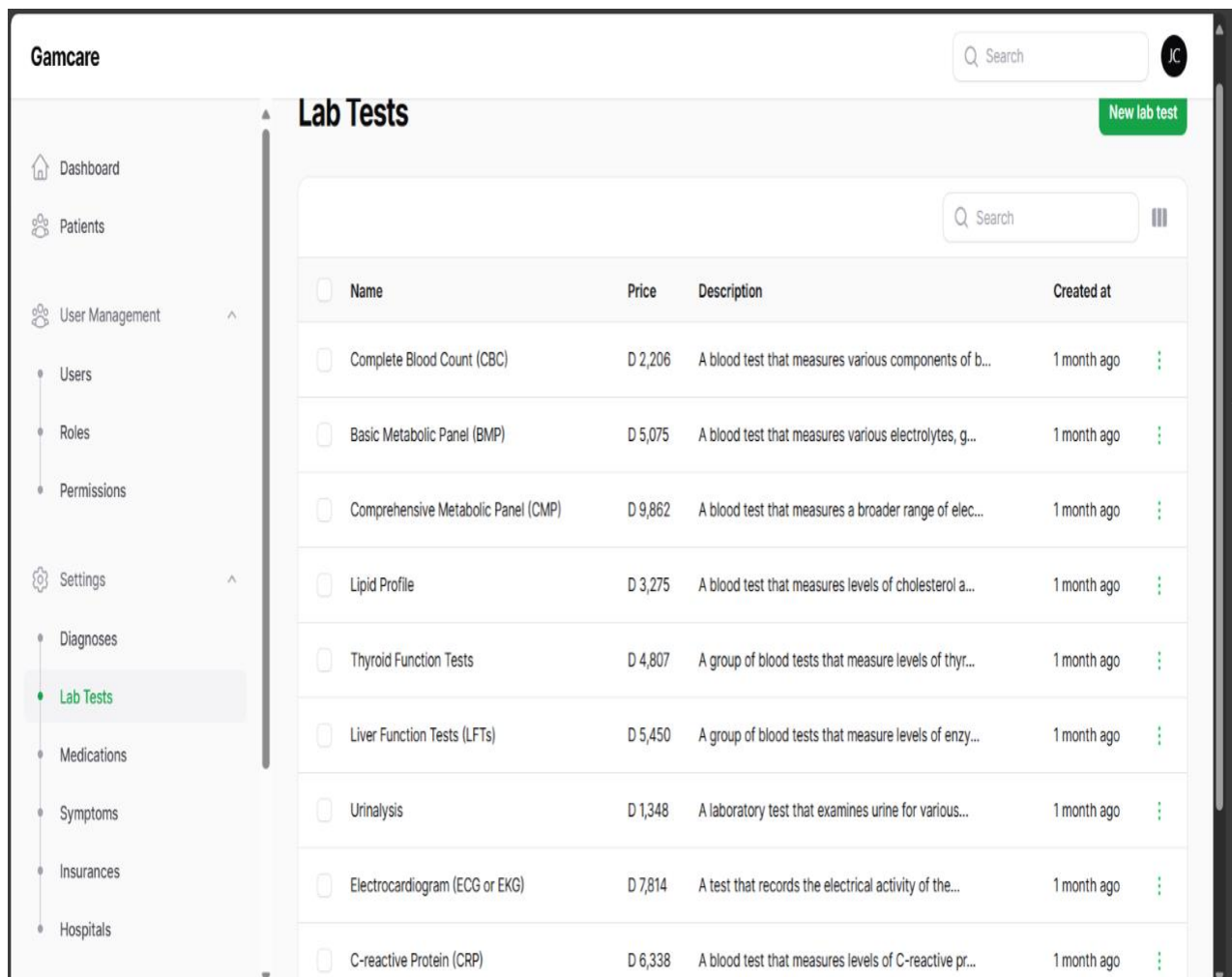


Figure 7: Lab test Page

This module handles all the payment in the system. Payments all are done in the cashier before doing the lap test. Following the nurse's assessment, the patient proceeds to see the doctor. The doctor reviews the information provided by the nurse and conducts a further examination. Based on the initial findings and patient's symptoms, the doctor may decide that additional laboratory tests are required for further diagnosis. If so, the patient is sent to the cashier to pay for the necessary laboratory tests, each lab test has a price attached to it. After payment is confirmed, the patient goes to the laboratory where samples and tests are conducted. The test results will be sent the doctor once they are ready and the patient returns to the doctor, who then reviews the test the results and prescribe the necessary medications.

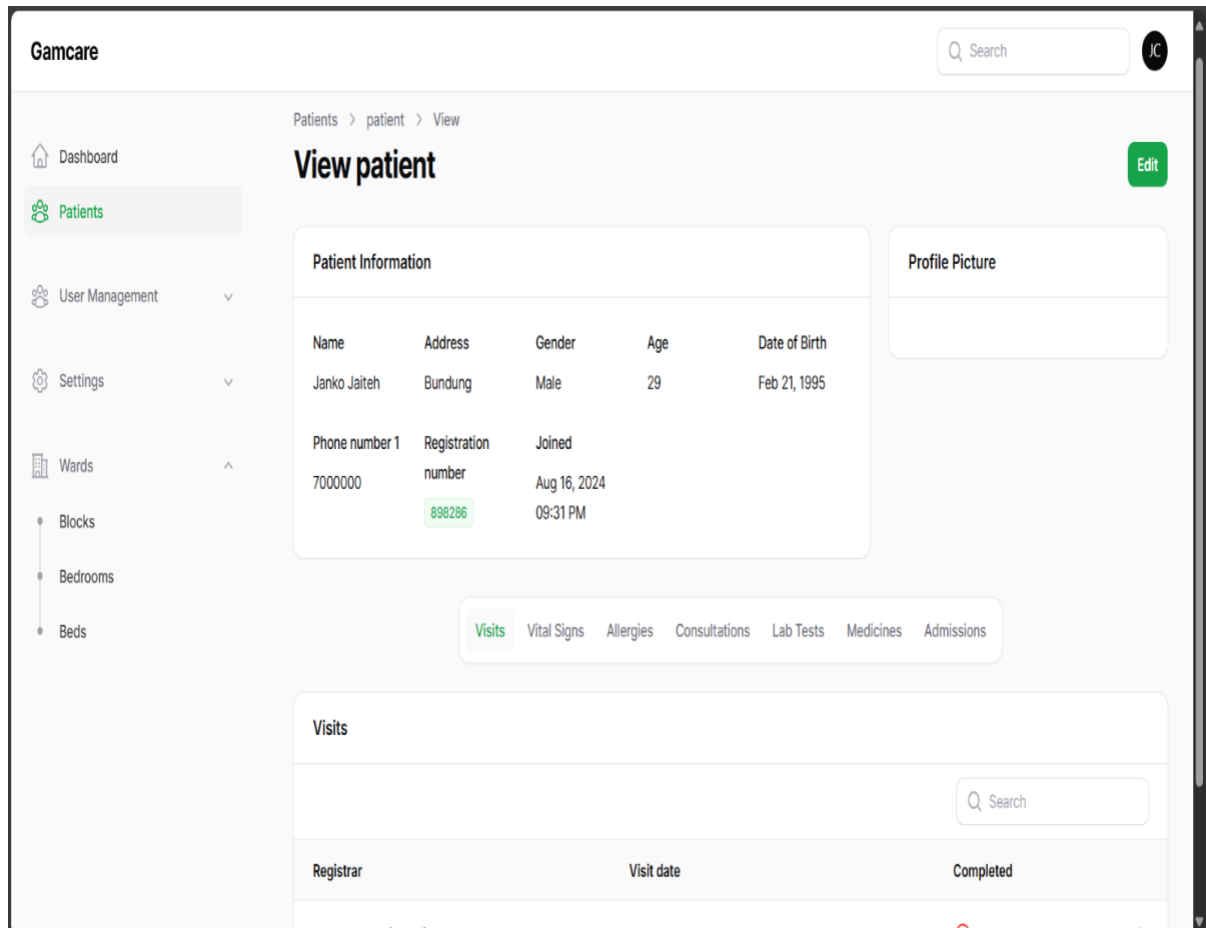


Figure 8: Patient's information Page

The patient information page is a user interface that displays the patient's data retrieved from the database. The patient's information page gives healthcare workers with a comprehensive overview of the patient's health records, recent visits, lab tests, consultations, medications and diagnoses. This page is important for doctors, nurses, and other healthcare workers, as it help them in making informed decisions based on the patient's pass medical records. when a patient file is not found, the doctors will not be able to look at the patient's history as a result they miss the previous diagnosis or medication of the patient. Once the patient's fingerprint is scanned, the system will automatically show up the patient's records in real time.

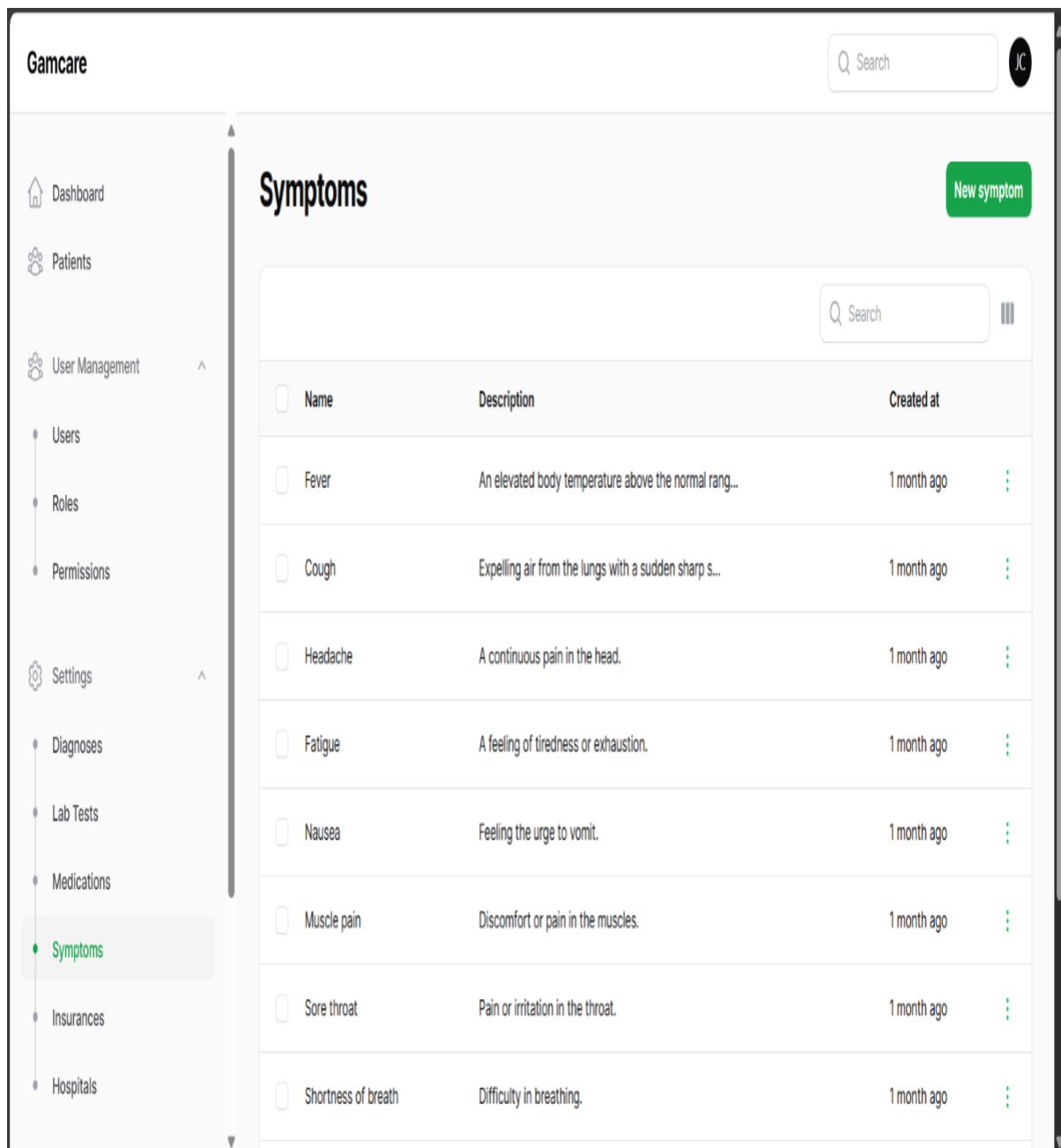


Figure 9: The patient's symptoms page

This page provides healthcare providers with patient's history and present health status.

The nurse and the doctor can add patients' symptoms in this page. This is where the nurse or the doctor documents the patient current health complains, symptoms and relevant observations. During the consultation, the nurse or the doctor inputs this information into the system which becomes patient past medical record.

Showing rows 0 - 4 (5 total, Query took 0.0003 seconds.)

```
SELECT * FROM `patients`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	id	slug	reference_number	name	gender	address	registered_at	profile_picture	dob	age	phone_number_1	phone_number_2	fee	email	blood_group
☐ Edit Copy Delete	1	jamal	611554	Jamal	Male	Bundung	1	NULL	2024-08-15	0	3424242	NULL	NULL	jamal@gmail.com	B+
☐ Edit Copy Delete	2	hamza	474743	Hamza	Male	Bundung	1	NULL	2024-08-31	0	2389281	NULL	NULL	hamzakante@gmail.com	A+
☐ Edit Copy Delete	3	samba-jallow	227358	Samba Jallow	Male	Jabang	1	NULL	1996-09-20	27	2268543	NULL	NULL	samba@gmail.com	B+
☐ Edit Copy Delete	4	pascal-sawneh	440287	Pascal Sawneh	Male	Bundung	1	NULL	2003-03-25	21	9283938	NULL	NULL	pascal@gmail.com	A+
☐ Edit Copy Delete	5	isatou-njie	675044	Isatou Njie	Female	Brusubi	1	NULL	2006-01-12	18	2220293	NULL	NULL	isatou@gmail.com	O+

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Figure 10: Patients table in the Hospital database

The hospital Database stores important information not just for their patients but the hospital as well in order to improve its efficiency.

Showing rows 0 - 24 (31 total, Query took 0.0003 seconds)

SELECT * FROM `medications`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

		id	name	price	description	created_at	updated_at
☐	Edit Copy Delete	1	Aspirin	649	A common pain reliever and anti-inflammatory medic...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	2	Ibuprofen	689	Used to relieve pain, reduce inflammation, and low...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	3	Acetaminophen	8216	A pain reliever and fever reducer.	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	4	Amoxicillin	173	An antibiotic used to treat bacterial infections.	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	5	Ciprofloxacin	220	An antibiotic used to treat various bacterial infe...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	6	Lisinopril	1080	An ACE inhibitor used to treat high blood pressure...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	7	Metformin	3402	Used to treat type 2 diabetes.	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	8	Simvastatin	744	A statin medication used to lower cholesterol and ...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	9	Loratadine	1584	An antihistamine used to relieve allergy symptoms.	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	10	Omeprazole	7182	A proton pump inhibitor used to reduce stomach aci...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	11	Atorvastatin	9001	A statin medication used to lower cholesterol and ...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	12	Gabapentin	2551	An anticonvulsant medication used to treat seizure...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	13	Albuterol	4932	A bronchodilator used to treat asthma and chronic ...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	14	Sertraline	3993	An antidepressant medication used to treat depress...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	15	Hydrochlorothiazide	7266	A diuretic used to treat high blood pressure and f...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	16	Metoprolol	8301	A beta-blocker used to treat high blood pressure a...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	17	Warfarin	926	An anticoagulant medication used to prevent blood ...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	18	Levothyroxine	7281	A synthetic thyroid hormone used to treat hypothy...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	19	Citalopram	1344	An antidepressant medication used to treat depress...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	20	Tramadol	459	A pain reliever used to treat moderate to moderate...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	21	Diazepam	7140	A benzodiazepine medication used to treat anxiety...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	22	Fluoxetine	611	An antidepressant medication used to treat depress...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	23	Amlodipine	669	A calcium channel blocker used to treat high blood...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	24	Metronidazole	7024	An antibiotic and antiprotozoal medication used to...	2024-08-30 22:40:20	2024-08-30 22:40:20
☐	Edit Copy Delete	25	Prednisone	3704	A corticosteroid medication used to treat inflamma...	2024-08-30 22:40:20	2024-08-30 22:40:20

Console

Figure 11: Medications table in the database

localhost

phpMyAdmin

Server: localhost - Database: gmcare - Table: roles

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers

Showing rows 0 - 7 (8 total, Query took 0.0003 seconds.)

SELECT * FROM `roles`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	id	name	guard_name	created_at	updated_at
☐ Edit ☐ Copy ☐ Delete	1	Super Admin	web	2024-08-30 22:40:17	NULL
☐ Edit ☐ Copy ☐ Delete	2	Admin	web	2024-08-30 22:40:17	NULL
☐ Edit ☐ Copy ☐ Delete	3	Doctor	web	2024-08-30 22:40:17	NULL
☐ Edit ☐ Copy ☐ Delete	4	Nurse	web	2024-08-30 22:40:17	NULL
☐ Edit ☐ Copy ☐ Delete	5	Pharmacist	web	2024-08-30 22:40:17	NULL
☐ Edit ☐ Copy ☐ Delete	6	Lab Technician	web	2024-08-30 22:40:17	NULL
☐ Edit ☐ Copy ☐ Delete	7	Cashier	web	2024-08-30 22:40:17	NULL
☐ Edit ☐ Copy ☐ Delete	8	Registrar	web	2024-08-30 22:40:17	NULL

Check all | With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Console

Figure 12: Users table in the database

Server: localhost - Database: gmcare - Table: diagnoses

1 | > >> | Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

			id	name	description	created_at	updated_at
☐	Edit	Copy	Delete	1	Common Cold	A viral infection of the upper respiratory tract c...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	2	Influenza (Flu)	A contagious respiratory illness caused by influen...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	3	Gastroenteritis	Inflammation of the stomach and intestines, often ...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	4	Migraine	A type of headache characterized by severe throbbi...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	5	Pneumonia	An infection that inflames the air sacs in one or ...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	6	Urinary Tract Infection (UTI)	An infection in any part of the urinary system, in...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	7	Allergic Rhinitis	An allergic reaction to airborne substances such a...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	8	Asthma	A chronic inflammatory condition of the airways ch...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	9	Sinusitis	Inflammation or swelling of the tissue lining the ...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	10	Acute Bronchitis	Inflammation of the bronchial tubes, usually cause...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	11	Hypertension (High Blood Pressure)	A condition in which the force of the blood agains...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	12	Diabetes Mellitus	A chronic metabolic disorder characterized by high...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	13	Coronary Artery Disease	A narrowing or blockage of the coronary arteries l...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	14	Gastric Ulcer	An open sore or lesion in the lining of the stomac...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	15	Anxiety Disorder	A mental health condition characterized by excessi...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	16	Depression	A mood disorder characterized by persistent feelin...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	17	Osteoarthritis	A degenerative joint disease characterized by the ...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	18	Rheumatoid Arthritis	An autoimmune inflammatory disease that affects th...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	19	Hyperthyroidism	A condition in which the thyroid gland produces to...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	20	Hypothyroidism	A condition in which the thyroid gland does not pr...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	21	Gout	A type of arthritis caused by the buildup of uric ...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	22	Chronic Obstructive Pulmonary Disease (COPD)	A group of progressive lung diseases, including em...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	23	Peptic Ulcer Disease	An open sore or lesion in the lining of the stomac...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	24	Obstructive Sleep Apnea	A sleep disorder characterized by repeated pauses ...	2024-08-30 22:40:20 2024-08-30 22:40:20
☐	Edit	Copy	Delete	25	Anemia	A condition characterized by a deficiency of red b...	2024-08-30 22:40:20 2024-08-30 22:40:20

1 | > >> | Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Figure 12: Symptoms table in the database

The Hospital Database is the database module that holds the records of patients, and the credentials of staff used in this system. The database serves as the central repository for all patient information. It stores patients' medical history, laboratory test results, and prescriptions. The database is also responsible for storing and managing fingerprint data, which is used for patient identification and retrieval of medical records.

There are two basic retrieval modes in the system which are: fingerprint and reference or ID number a six-digit number that is generated to every user in the system.

CHAPTER FOUR

4.0 SUMMARY, CONCLUSION AND RECOMMENDATIONS

4.1 Summary

Across numerous African nations including the Gambia, healthcare facilities depend heavily on manual system for documenting and retrieving patient information. Medical information is the most sensitive and valuable information in terms of privacy which implies all information related to the patient. A robust and effective medical information system should be made up of storing, editable, and making available all clinical information to the doctors and staff for patient care (Kalaiselvan & Bahari, 2021). These paper-based system is prone to errors, inefficiencies and delays in patient care which can affect the overall healthcare delivery. Patient records are stored manually in a logbook or files which is frequently misplaced, vulnerable to loss damage, unauthorized access or even loss which can lead to difficulties in tracing patient medical history.

If a file is not found in time, a new file is created for the patient, which occasionally results in neglecting a patient allergies or prior conditions. These challenges result in complications where most clinicians do not know the history of their patients and therefore give false diagnoses and wrong medications to patients without knowing the medical history of each patient (Addison Kobla Azameti et al., 2022).

The problem in most hospitals in the country is that they cannot match the patient with their prior files, files are mostly sorted on the last name and most of the people share the same last name in the Gambia. This leads to difficulties in tracking patient history and diagnosis in ensuring continue patient care.

To address these challenges there must be a more dependable and efficient method of managing patient data. The biometric fingerprint registration and retrieval systems offer a promising solution to address these challenges.

Having electronic health records has helped hospitals reduce paperwork and alleviate the shortage of healthcare workers. As more and more industries embrace the features of ICTs to ensure the way they serve their customers, the healthcare industry is largely affected because effective or ineffective customer can lead to permanent impairment or death (Otuagoma et al., 2023)

The Gamcare Hospital Management provides a unique and secured way to identify and authenticate individuals, making it the best technology for patient information management system. The system allows for a quick and accurate retrieval of patient records which provides healthcare providers with access to complete and up-to-date information at the point of care. The system would help in reducing duplication of records as each patient's fingerprint is linked to a unique identifier, this will prevent the creation of multiple records for the same individual and it will reduce the likelihood of human errors in keeping records.

We must reduce the time it takes to file, retrieve and maintain patients records and reduce the space these records displace and ensure that we protect our records from premature destructions.

4.2 Conclusion

This research examines the challenges facing the hospitals in terms of patient data management and to come up with solutions in mitigating these challenges.

Based on both observations and interviews in the hospital, the patient information management system in hospitals in the Gambia is labor-intensive. The manual patient information system which relies heavily on physical paper records, has several inefficiencies such as delays in accessing patient data, misplaced, human errors and difficulties maintaining continuity of care when records are misplaced. The huge influx of patients visiting the clinic makes the process of using paper-based records keeping extremely difficult. Many research papers indicate the need to improve the paper-based management system (Addison Kobla Azameti et al., 2022). The current system is time-consuming when it comes to information retrieval, organization, and filing. It also has poor security, because access to information cannot be monitored correctly.

The Gamcare Hospital Management system provides a unique and secured approach to identifying and verifying individuals, making it the best the technology for patient management. The system will reduce the time it takes to register new patients, retrieve their records and improved the overall efficiency of the hospital by reducing the delays and waiting times. Furthermore, the system assists in helping in mitigating the duplication of records by linking each patient's fingerprint to a unique identifier, this will prevent the creation of multiple records for the same individual and it will reduce the likelihood of human errors in keeping records. The system focuses mainly on patient care, laboratory, admin and pharmacy issues, and how to store and retrieve data from the systems.

4.3 Limitations

The collection of data from the hospital was challenging due to difficulties in reaching the top officials of the hospital and the sensitive nature of medical data which involves strict security and privacy regulations.

4.4 Future works

The future development of this system will be focusing on the following:

- The development of a mobile application for this system would allow patients to be reminded about their prescriptions details, upcoming appointments and other relevant information the patient needs to be reminded about. The automated appointment reminder in the mobile appointment scheduling system also saves time as hospital operators wont required to call and send SMS to patients reminding them of their appointments. This will enhance the efficiency of the hospital management system and provide convenience to patients (Kyambille & Kalegele, 2015).
- Development of an internet version of the system that will enable a secure transfer of patient records among hospitals in The Gambia with the use of internet connection.

4.5 Recommendations

- I. **Stakeholder Engagement and Collaboration:** Engagement with relevant stakeholders such as the government, non-government organization (NGOs), healthcare providers and patient about the benefits of the system. Patients are defined as one of the crucial stakeholders of health care and decision-making, and this shows the need for involving them in the treatment process (Marzban et al., 2022). Public awareness should also be encouraged these can be in a form of radio Campagne, workshops, television talk shows, newspapers etc. These public engagements will help in bridging the gap between the healthcare professionals, patients and the general public at large with the aim of working hand in glove in improving the health sector.
- II. **Staff Training and Capacity Building:** For healthcare personnel to use the biometric system more efficiently they must receive proper training. This includes not only training them on how to use the system but to also understanding the importance of data accuracy and security with regards to health. Biometric systems gather sensitive personal information, and any breaches could significantly impact patient trust and wellbeing. Health data contains sensitive private information's which includes the identity of patient, medical conditions of the patient etc. Leakage

of these private information affects the personal life, including bullying, and loss of job due to medical history (Thapa & Camtepe, 2021).

- III. **International Fundings and Grant:** Seeking fundings from international organizations like World Health Organization (WHO) and world Bank will help in the procurement of the necessary equipments for the implementation of the biometric patient information management. The funding also will help in the maintenance of the system and will cover the expenses involve in the training of healthcare providers.

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